Akoizo Original Practice - Form 4

Division C | Electric Vehicle | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) A proportional controller sets motor power based on:

Unit: Control: speed profiles, stopping, and hitting the Target Time.
Worked-case data: Total MP travel is 8.60 m forward plus 1.60 m back. Target Time is 15.0 s. You allow 1.0 s for each of four speed ramps, 0.5 s to reverse, and 0.3 s of settling. What cruise speed is needed?
Separate investigation observations:
Fresh batteries: Fixed-power run: 13.4 s, stop error 4 cm.
Half-drained: Same code: 15.9 s, stop error 9 cm.
Closed-loop version: Both battery states: 15.0 ± 0.2 s, stop error 1-2 cm.

A. The elapsed time only

B. The pusher width

C. The remaining distance to the target

D. The battery's label

Answer: _____

2. (2 points) Target Distance 6.00 m and Bottle Line Distance 1.50 m. Where is the Bottle Line from the Start Point?

A. 1.50 m

B. 6.00 m

C. 7.50 m

D. 4.50 m

Answer: _____

3. (2 points) Which action results in a competition violation?

Unit: Competition day: impound, check-in, and running.
Worked-case data: With 1:30 left in event time, a team's first run was successful. Its vehicle now has a loose wire. What should it do?
Separate investigation observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

A. Following the vehicle down the track before being called

B. Measuring the track during event time

C. Cleaning the track while keeping it dry

D. Programming over USB

Answer: _____

4. (2 points) A pusher's widest opening is 35.0 cm. What is its Pusher Bonus?

A. 0

B. -52.5

C. +35

D. -35

Answer: _____

5. (4 points) Why calibrate distance per count with a measured run?

Unit: Drivetrain and distance sensing: motors, gears, and encoders.
Worked-case data: A 6.5 cm wheel drives a 100-count-per-revolution encoder through a 3:1 motor-to-wheel gear ratio, with the encoder on the motor shaft. How many counts for 8.60 m?
Separate investigation observations:
Forward test: Commanded 500 cm, measured 503.5 cm.
Reverse test: Commanded 150 cm back, measured 148.9 cm.
Tires: Rubber O-rings, slightly compressed under load.

Answer: _____
_____
_____

6. (4 points) A vehicle pauses 2 s after the pencil trigger before moving. Does the pause count?

Answer: _____
_____
_____

7. (4 points) Describe a good Measurement Point.

Unit: Power, construction limits, and legal parts.
Worked-case data: A team's vehicle uses six NiMH AA cells, a motor driver chip, a microcontroller, a 1 F supercapacitor to boost acceleration, and a laser pointer for aiming. What must change?
Separate investigation observations:
Parts list: 8 x AA alkaline, Arduino, encoder, 12 V boost converter.
Inspection question: Does the boost converter store energy to exceed battery power?
Team explanation: It only steps up voltage continuously from the batteries; no large storage.

Answer: _____
_____
_____

8. (4 points) Why test at extreme combinations of posted values?

Unit: Control: speed profiles, stopping, and hitting the Target Time.
Worked-case data: Total MP travel is 8.60 m forward plus 1.60 m back. Target Time is 15.0 s. You allow 1.0 s for each of four speed ramps, 0.5 s to reverse, and 0.3 s of settling. What cruise speed is needed?
Separate investigation observations:
Fresh batteries: Fixed-power run: 13.4 s, stop error 4 cm.
Half-drained: Same code: 15.9 s, stop error 9 cm.
Closed-loop version: Both battery states: 15.0 ± 0.2 s, stop error 1-2 cm.

Answer: _____
_____
_____

9. (4 points) How does sideways drift affect the score even if the forward distance is perfect?

Answer: _____
_____
_____

10. (4 points) Why is testing on the competition floor beforehand restricted?

Unit: Competition day: impound, check-in, and running.
Worked-case data: With 1:30 left in event time, a team's first run was successful. Its vehicle now has a loose wire. What should it do?
Separate investigation observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: Compare pushers with openings of 9.0 cm and 15.0 cm. The 9.0 cm version catches the bottle in 8 of 10 tests; the 15.0 cm version in 10 of 10. A miss costs about 420 points (400 cm bottle penalty plus no bonus).

Answer: _____
_____
_____

12. (2 points) What should the team do?

Supplied synthetic observations:
Forward test: Commanded 500 cm, measured 503.5 cm.
Reverse test: Commanded 150 cm back, measured 148.9 cm.
Tires: Rubber O-rings, slightly compressed under load.

A. Switch to time-based stopping

B. Use separate calibration factors for forward and reverse

C. Ignore the error because it is under 1 percent

Answer: _____

13. (2 points) Define “Measurement Point (MP)” in the context of Electric Vehicle.

Answer: _____
_____
_____

14. (2 points) Define “AA battery” in the context of Electric Vehicle.

Answer: _____
_____
_____

15. (2 points) Define “Closed-loop control” in the context of Electric Vehicle.

Answer: _____
_____
_____

16. (2 points) Define “Bottle Location” in the context of Electric Vehicle.

Answer: _____
_____
_____

17. (2 points) Define “Ready-to-run” in the context of Electric Vehicle.

Answer: _____
_____
_____

18. (2 points) Define “Pocket” in the context of Electric Vehicle.

Answer: _____
_____
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